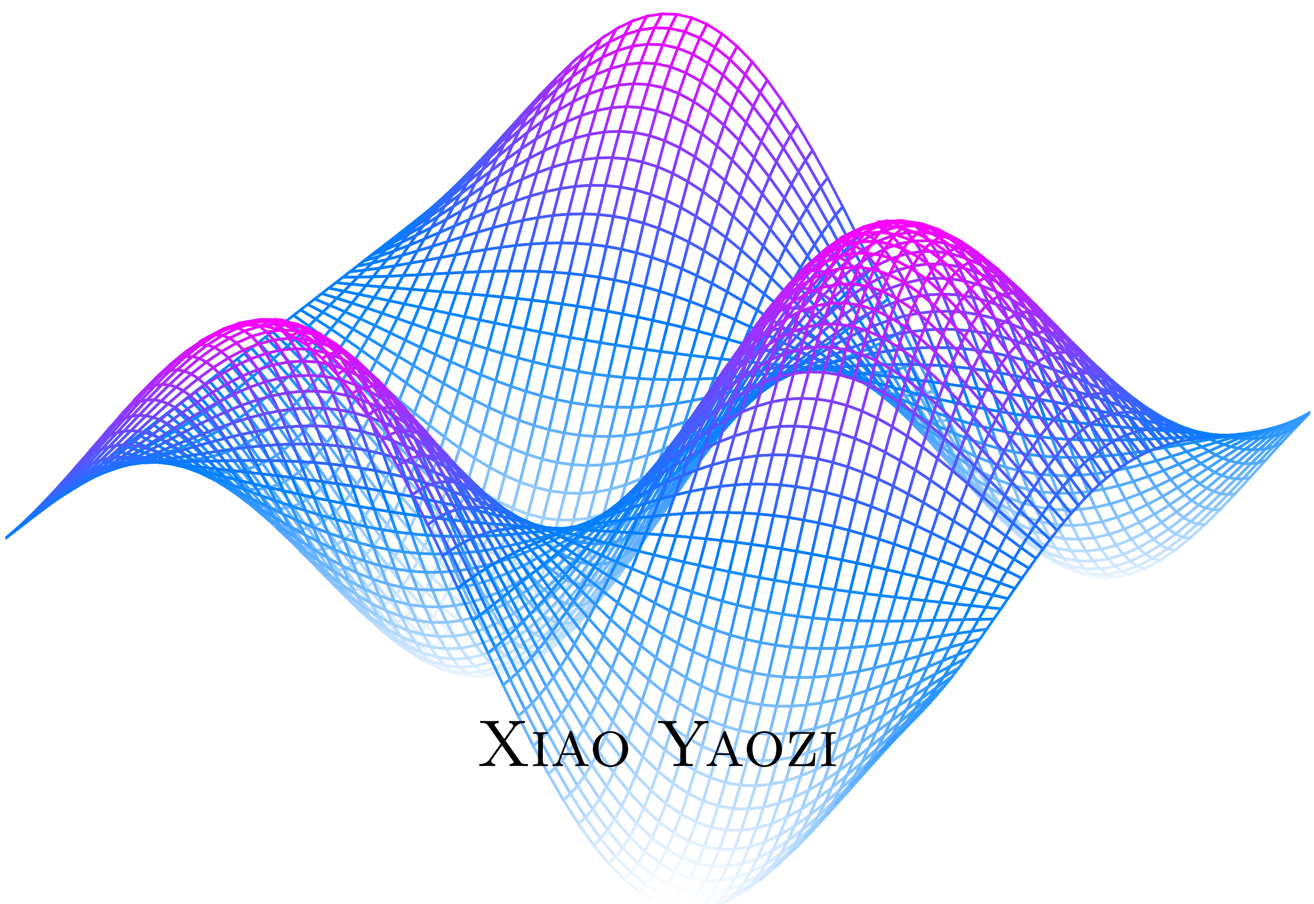
Practical Calculus

FIRST EDITION



Xiao Yaozi

Preface to the first edition

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Contents

PART

I

Precalculus

CHAPTER 1

Functions

SECTION 1.1

Function basics

Before we start on the calculus topics, this part of the book will acquaint you with the mathematics you would need to understand in order to read this book. If you are familiar with the these topics, you may skip to the Part II on Calculus I. For the rest of you, the pace of this part will be very fast, it is recommended that you read up on the concepts you do not understand very well. We will start with the idea of a *function*. By now, I am sure that many of you may have seen equations like these:

$$x^2 - 5x + 6 = 0 \tag{1.1}$$

$$\sin(x) = 0.5, 0 \leq x \leq \frac{\pi}{2} \tag{1.2}$$

$$y = mx + c \tag{1.3}$$

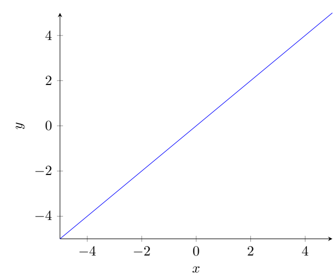


Figure 1.1. Graph of y=x.

Out of the three equations, notice that Equation (1.3) is special. There is not some fixed x values that you must plug in. Instead, the y value varies with the x value. Given a x value, the equation "produces" a y value. This brings us to the idea of a function. A function is simply something that takes in a value, and "produces" another value based on the input. You may have came across functions before, such as the trigonometric functions like sin and cos. The value of $\sin(x)$ varies with different values of x. Of course, it is time for me to give you the definition of a function.

Definition 1.1

A function f is a mathematical object that assigns its inputs, or *domain*, to its outputs, or *range*. The notation $f : A \to B$ means that the function has a domain A and range B. The notation $f(a)$ where $a \in A$ denotes the element of B to which a is assigned to.

Note that A and B are sets.

Example

Here are some examples of functions:

$$f(x) = x + 1 \tag{1.4}$$

$$g(x) = \sin(x) \tag{1.5}$$

$$h(x) = x^2 - 5x + 6 \tag{1.6}$$

To evaluate the output of a function, you simply substitute the input to the expression that defines the function. Notice that for Equation (1.4), it is simply the line $y = x + 1$. Each value of $f(x)$ for some $x \in \mathbb{R}$ is just y. Take $x = 1$ for example, $f(1) = 1 + 1 = 2$ and for the line $y = x + 1$, notice the y value here is also 2. The function assigns each x value to $x + 1$. Hence, in this case, $f : \mathbb{R} \to \mathbb{R}$. For Equation (1.5) , the function takes any real x value and maps it to $\sin(x)$. As a result, its domain would be $\mathbb{R}$ and its range would be $[-1, 1]$. Again, it is just like the graph $y = \sin(x)$.

Lastly, the quadratic in Equation (1.6) is another real function(i.e. it has a real domain and range). Think about what happens when $h(x) = 0$, what value of x must you plug in to make $h(x) = 0$. Well, isn't it just the roots of the quadratic. Hence, the roots of a function are just the values x such that $h(x) = 0$. In this case, it would be 2 and 3. Thus, $h(2) = h(3) = 0$.

Actually, functions can be more than the usual $f(x) =$ something. Let's say you want to have a function that outputs "1" if the input is rational and "0" if it is not, you can define it as such:

This is actually the famous Dirichlet function that shatters certain intuitions on functions and will be discussed in later chapters.

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \notin \mathbb{Q} \end{cases} \tag{1.7}$$

Here, $f(\frac{1}{2})$ would be 1 but $f(\sqrt{2})$ would be 0. This is an example of having conditionals in a function. Lastly, functions may take more than one input and produce more than one output, examples include:

$$f(x, y) = x^2 + y^2 \tag{1.8}$$

$$g(t) = (\cos(t), \sin(t)) \tag{1.9}$$

Equation (1.8) is an example of a *multivariate* function, in other words, it takes in multiple inputs. If $x = 2$ and $y = 2$, then $f(2, 2) = 8$. On the other hand, Equation (1.9) does not have a single output. Instead, it has a tuple as its output. Evaluating it at $t = \pi$, notice that the output is $(-1, 0)$. What is interesting is that if you trace out its outputs on a plane for all $t \in \mathbb{R}$, you will get:

As you will see in Part IV of the book, this is the parametric equation of a circle.

A tuple is basically a set where order matters. You can treat it as if it is a coordinate.

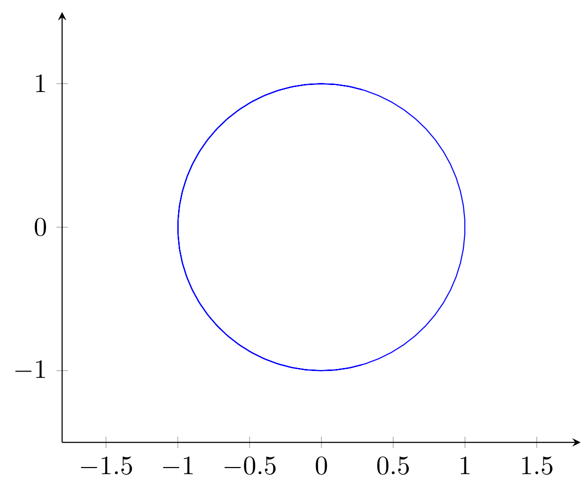


Figure 1.2. Plot of Equation (1.9) for all $t \in \mathbb{R}$

Functions are the building blocks of mathematics, and this book will deal extensively with it. Therefore, it is paramount that you get the basics right. Try the following exercise:

Exercise

1. Given that $x = 2$ and $y = \pi$, evaluate $f(x)$ for:
   (a) $f(x) = x^3 + 1$
   (b) $f(x) = 2^x$
   (c) $f(x) = \begin{cases} x^4 + 1 & \textit{if } x < 2 \\ 3^x & \textit{otherwise} \end{cases}$

(d) $f(x, y) = \frac{\sin(y)}{x}$

2. Given the following functions, write down $f : A \to B$ where A is the domain and B is the range(do not consider complex inputs and outputs):

   (a) $f(x) = x$

   (b) $f(x) = \frac{1}{x}$

   (c) $f(x) = \sqrt{x}$

This function has a special name, the ***identity*** *function. Notice how its input and output are the same.*

3. Find the root(s) of the following functions:

   (a) $f(x) = x^2$

   (b) $f(x) = 2^x - 1$

   (c) $f(x) = \sin(x)$

Solution

1. (a) 9

   (b) 4

   (c) 9

   (d) 0

2. (a) $f : \mathbb{R} \to \mathbb{R}$

   (b) $f : \mathbb{R} \setminus \{0\} \to \mathbb{R} \setminus \{0\}$ (Note the function is not defined at $x = 0$)

   (c) $f : \mathbb{R}_0^+ \to \mathbb{R}_0^+$ (Square root not defined for negative numbers)

3. (a) 0

   (b) 1

   (c) $n\pi, \forall n \in \mathbb{Z}$

SECTION 1.2

Function composition and inverses

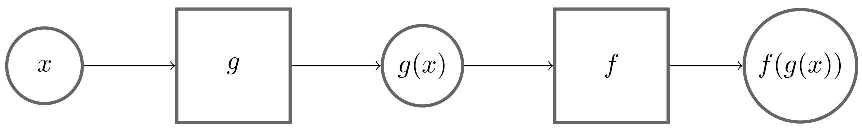
In this chapter, we will look at composition of functions and inverse functions. You can use the normal arithmetic operations on functions like how you would to numbers:

Example If $x = 2$ and $f(x) = x^2 + 4$:

$$f(x) + f(x) = 2^2 + 4 + 2^2 + 4 = 16$$

$$\frac{f(x)}{2} = \frac{2^2 + 4}{2} = 4$$

What is more interesting, perhaps, is taking the *output* of a function and use it as the *input* to another function. You may also think of it as "piping" one function into another. This is called *function composition*.



Note that the rectangles are functions and circles are values.

Figure 1.3. Visual representation of function composition.

Another notation for composing functions together(which we will not really use) is as such:

$$f(g(x)) = (f \circ g)(x)$$

When you compose the same function n-times, the notation for it is:

$$f^n(x) = \underbrace{f(f(\ldots f(x)))}_{n-times}$$

However, this does not apply to trigonometric functions. $\sin^2(x)$ *is* $\sin(x) \times \sin(x)$ *instead of* $\sin(\sin(x))$.

Example Here are some examples of composing functions together: If $x = 3$ and $f(x) = \pi x$, $g(x) = \sin(x) + 1$:

$$g(f(x)) = \sin(\pi \times 3) + 1 = 1 \tag{1.10}$$

$$f^3(x) = \pi^3 x \tag{1.11}$$

There is nothing much here, really, just evaluate the function from the innermost function to the outermost.

Theorem 1.1 Given functions $f : X \to Y$, $g : Y \to Z$ and $h : Z \to W$, prove that the composition of the three functions is *associative*, in other words:

$$h \circ (g \circ f(x)) = (h \circ g) \circ f(x)$$

PROOF Let $x \in X$, we have:

$$h \circ (g \circ f(x)) = h(g \circ f(x)) = h(g(f(x))) = h \circ g(f(x)) = (h \circ g)f(x)$$

This proves the associativity of function composition. QED

Remark This is just a fun little exercise for the reader and ultimately is not that important to the entire book, just a good identity to keep in the back of your mind.

Next, we will delve into inverse functions. Let us start with one I believe you may have seen before, look at the following figure:

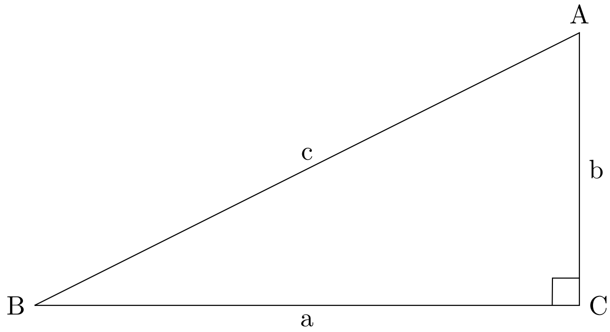


Figure 1.4. Figure of right triangle ABC.

If I ask you what is $\arcsin(B)$, you may say it is $\frac{b}{c}$. But, what is $\arcsin(\sin(B))$, well, that is just B, isn't it. Hence, the arcsin sort of "cancels" the sin. We call arcsin the *inverse function* or just inverse of sin. Generally, given a function $f : X \to Y$, f^{-1} denotes the inverse function that satisfies:

We will use arcsin *as the inverse of* sin *instead of* $\sin^{-1}$ *as it is clearer.*

$$f \circ f^{-1}(y) = y, y \in Y \text{ and } f^{-1} \circ f(x) = x, x \in X \tag{1.12}$$

One common misconception is that $f(f^{-1}(x)) = f^{-1}(f(x))$. While this is true for many functions like $f(x) = x + 1$, it is *not* always true. For example, take the following function:

$$f(x) = \sin(x), f : \mathbb{R} \to \{y : -1 \leq y \leq 1\}$$

This looks deceptively simple, when asked to evaluate $f^{-1}(f(\pi))$, it is π (You may want to verify it yourself). But what is $f(f^{-1}(\pi))$? Well, according to our definition of f, $f^{-1}(\pi)$ is simply not defined. In actuality, to find $f^{-1}(\pi)$, we need complex numbers, which is not allowed in the definition. Hence, you must be extra careful when you are dealing with function inverses and pay attention to the domain and range.

To find the inverse function of a function f, we have to find the function that maps the outputs of the f back to its inputs. To solve algebraically, we set $f(x)$ for some arbitrary x to y, and make x the subject of the equation.

Example

Given that $f(x) = \frac{x+7}{3}$:

$$\begin{aligned} f(x) &= y \\ \frac{x+7}{3} &= y \\ x + 7 &= 3y \\ x &= 3y - 7 \end{aligned}$$

Hence, the inverse function is $f^{-1}(y) = 3y - 7$.

Example

Some common examples of inverse functions:

1. Trigonometry:

   arcsin arccos arctan ...

2. Logarithms (Inverses of exponential functions):

   ln $\log_b$

Exercise

1. Given $f(x) = \frac{x^2+3}{2}$ and $g(x) = \tan(x) + 1$, evaluate:
   (a) $f(g(\frac{\pi}{4}))$
   (b) $f^3(1)$
   (c) $f(g(\arcsin(0)))$
2. Given that $h(x) = (x-7)^2$, $x \geq 7$, find the inverse of h.
3. Hence, evaluate:
   (a) $h^{-1}(1)$
   (b) $h(h^{-1}(69))$

Solution

1. (a) $\frac{7}{2}$
   (b) $\frac{61}{8}$
   (c) 2
2. $h^{-1}(y) = \sqrt{y} + 7$
3. (a) 8
   (b) 69 (Inverse function and the original function "cancels".)

SECTION 1.3

Injection, surjection and bijection

You may have noticed in the previous exercise that question 2 is particularly interesting. There is a restriction on the domain of $h(x)$. Why is that? This is what this chapter is about. However, before we explain that, I will go over some definitions of properties of functions. Firstly, we have *injection.*

Definition 1.2 An injective function, f, or a *one-to-one* function, is a function such that every distinct output of the function corresponds to only *one* distinct input, that is:

$$f(x_1) = f(x_2) \text{ implies } x_1 = x_2$$

By contraposition, $x_1 \neq x_2$ implies $f(x_1) \neq f(x_2)$. The contraposition of a statement "p then q" is "not q then p".

Of course, this explanation might not have been very clear. Here is a visualisation:

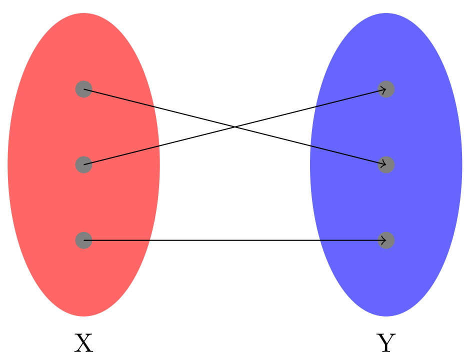


Figure 1.5. Illustration of an injective function with domain X and range Y.

As you can see in Figure 1.5, every output in Y of the function is associated with only one input in X, hence the function is *injective.* To prove that a function f is injective, you can either show that if $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$, or if $f(x_1) = f(x_2)$, then $x_1 = x_2$.

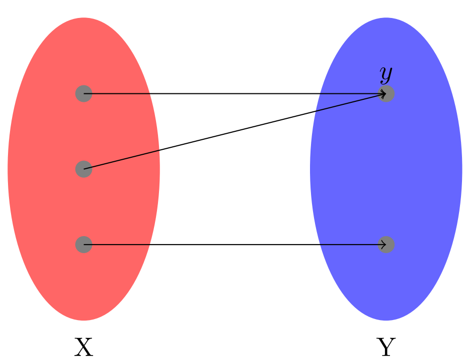


Figure 1.6. Illustration of a non-injective function with domain X and range Y.

Here, the function is *not* injective. As you can see in Figure 1.6, the output y is associated with more than one input in X. With this in mind, it is time to talk about *surjection.*

Definition 1.3 A surjective function f is a function such that for every output y, there exists at least one input x such that $f(x) = y$.

From Figure 1.5 and Figure 1.6, both functions are surjective as every output is associated with at least one input. Do take note that it does not matter how many inputs associate to one output, as long every output is associated with at least one input, the function is surjective. To prove that a function is surjective, you must show that there exists an element in the domain of the function for any arbitrary output in the range.

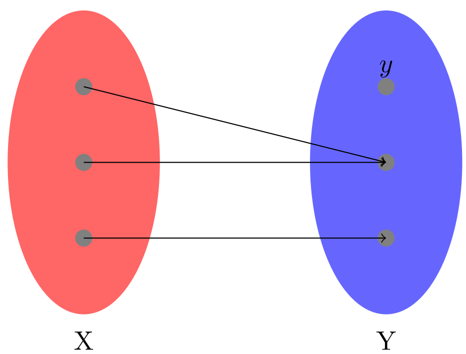


Figure 1.7. Illustration of non-surjective function with domain X and range Y.

From Figure 1.7, we can see that y has no input associated with it. As a result, the function here is not surjective. Lastly, *bijective* functions can't be any easier.

Definition 1.4 A bijective function is a function that is both injective and surjective.

However, bijective functions have a superpower. A function is only bijective *if and only if* it has an inverse.

P iff Q, or "P if and only if Q", means that "if P then Q" and "if Q then P".

Theorem 1.2 A function, $f : A \to B$, has an inverse if and only if it is bijection.

PROOF To begin such proofs where there is a "if and only if" clause, we will prove both cases separately. First, we claim that if the function has an inverse, then it is bijective. To prove this, we simply prove that f is injective and surjective (See Lemma 1.1). Then, we claim that if a function is bijective, then it has an inverse (See Lemma 1.2). With both proofs, we show that f has an inverse if and only if it is a bijection. QED

Lemma 1.1 If a function, $f : A \to B$, has an inverse, then it is bijective.

PROOF Suppose f^{-1} is the inverse of f. Notice that $f^{-1}(f(x))$ is injective as it is simply $f(x) = x$. Suppose $a, a' \in A$ and $f(a) = f(a')$, then $f^{-1}(f(a)) = f^{-1}(f(a')))$. Notice that by injectivity of $f^{-1}(f(x))$, $a = a'$. Hence, f is injective. Then, suppose $b \in B$, we wish to show that the function is surjective. Also, $f(f^{-1}(x))$ is surjective(it is the identity function). Let $f^{-1}(b) = x$ and $f(f^{-1}(b)) = c$ for some $c \in B$. By a simple substitution, $f(f^{-1}(b)) = f(x) = c$, thereby showing there exists a $x \in A$ for every $c \in B$. The two cases proves that f is bijective if it has an inverse. QED

Lemma 1.2 If a function, $f : A \to B$, is a bijection, then it has an inverse.

PROOF Let us define f^{-1} as $f^{-1} : B \to A$. Let $b \in B$. Since f is surjective, there exists $a \in A$ such that $f(a) = b$. Let $f^{-1}(b) = a$. Since f is injective, f^{-1} is well-defined, or a function that outputs the same value given the same input. Next, we will show that f^{-1} is indeed the inverse of f by showing the composition of the two is an identity function. Again, let $a \in A$ and $b = f(a)$ which implies, by definition, $f^{-1}(b) = a$. Composing the two functions, $f(f^{-1}(b)) = f(a) = b$, which shows that f^{-1} is indeed the inverse function. The other case of $f^{-1} \circ f$ is similar. Hence, if f is a bijection, then it has an inverse. QED

Remark This serves as a very good example showing how the reader should prove the injectivity, surjectivity, and bijectivity of functions. Also, this explains why in the previous exercise, the function $h(x) = (x-7)^2$ is bounded, only taking x values larger than 7. The function is not bijective in $\mathbb{R}$, as evident in the graph as it is not injective. Hence, the bounds are needed for h to have an inverse.

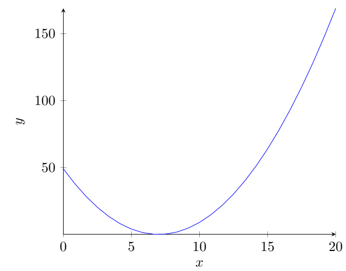


Figure 1.8. Plot of $(x-7)^2$

Exercise

1. Determine if the following functions are injective or surjective or bijective in its domain:
   (a) $f(x) = \sqrt{x}, f : \mathbb{R}^+ \to \mathbb{R}^+$
   (b) $f(x) = \frac{1}{x}, x > 0$
   (c) $f(x) = (x-7)^2, x \geq 5$
2. *Prove* that:
   (a) The composition of two injective functions is *injective*.
   (b) The inverse of a function is *injective*.
   (c) The inverse of a function is *surjective*.

Solution

1. (a) Bijective
   (b) Injective
   (c) Surjective
2. (a) Let the two functions be $f : A \to B$ and $g : B \to C$. If $g(f(x_1)) = g(f(x_2)))$, by injectivity of g, $f(x_1) = f(x_2)$ and by injectivity of f, $x_1 = x_2$. Thus, $g \circ f$ is injective.
   (b) Let $f : A \to B$ and f^{-1} be its inverse. Notice if $f^{-1}(x_1) = f^{-1}(x_2)$, then $f(f^{-1}(x_1)) = f(f^{-1}(x_2))$ which implies $x_1 = x_2$. Hence, f^{-1} is injective.
   (c) Similarly, define f and f^{-1}. Let $y \in A$ and $f^{-1}(x) = y$. Notice that $x = f(y)$ and because f is surjective(by Lemma 1.1), $x \in B$, thus proving that f^{-1} is surjective.

And this concludes the chapter on functions. Readers are encouraged to do more similar practices to strengthen their knowledge on functions. Also, this chapter is more on the "pure mathematics" side, such proofs are only presented as a good exercise to the reader. In actuality, the book does not focus that much on this kind of mathematics(i.e. analysis) but more on the practical aspects of Calculus.

Section 1.4

An aside: A more rigorous definition of a function

TBD

CHAPTER 2

Linear Algebra

SECTION 2.1

Vector Basics

Firstly, good job on finishing chapter 1, it is not an easy feat. Stepping into this chapter, you may feel overwhelmed by the name, *Linear Algebra*. What is that? Please take note though, the knowledge in this chapter will not be used until Calculus III to IV in part IV of the book. If you are just willing to learn Calc I and II, you do not need the knowledge here. With that said, let's delve into Vectors. A concept that is very ubiquitous in this world is a coordinate, like the GPS coordinate in the maps app. A coordinate simply describes the position of a point in space. Generally, it is denoted (x, y) in 2 dimensions and (x, y, z) in 3 dimensions. Likewise, vectors are also behaves like coordinates and are written down in a *similar* fashion. Think of a vector as a coordinate.

Definition 2.1 A vector is a mathematical object with a magnitude, or length, and a direction. A vector, $\vec{a}$ is a point (x, y) in space drawn with an arrow. It is denoted as:

$$\vec{a} = \begin{bmatrix} x \\ y \end{bmatrix}$$

Likewise, a vector at (x, y, z) is denoted $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$

Of course, vectors by themselves are not very useful. Hence, we shall define the vector space. For the purposes of this text, you may think of vector spaces as the 2 and 3 dimensional euclidean spaces you are very familiar with. In a vector space, we can finally meaningfully manipulate them. Take a look at the vector $\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and the vector $\vec{b} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$ in the two dimensional euclidean space:

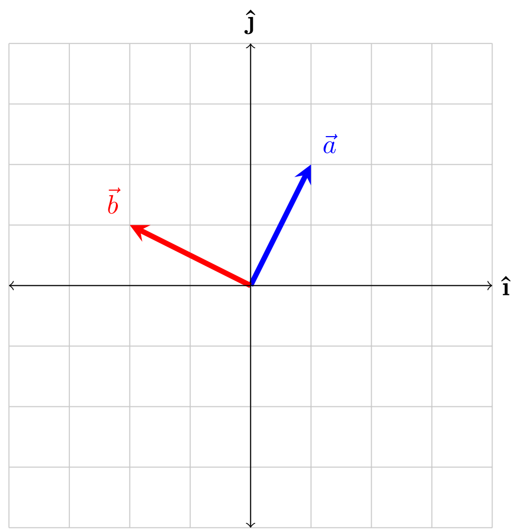


Figure 2.1. Figure of vectors $\vec{a}$ and $\vec{b}$.

You may have noticed that the coordinate axes in Figure 2.1 are unusual. Instead of the usual xy axes, you have the $\hat{\imath}$ and $\hat{\jmath}$ axes. This is a convention in linear algebra. $\hat{\imath}$

Some authors use $\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}}$ instead of the usual $\hat{\imath}, \hat{\jmath}, \hat{k}$. We shall use the former.

translates to the x axis and $\hat{\mathbf{j}}$ translates to the y axis. Likewise, in the third dimension, $\hat{\mathbf{k}}$ represents the z axis. Another thing to note are the *basis vectors*:

Definition 2.2

The set of vectors in a vector space are called *basis* if every vector in the vector space can be written uniquely written as a finite *linear combination* of the basis vectors. An example would be these vectors $\hat{\mathbf{i}} = \left[\begin{smallmatrix}0\\1\end{smallmatrix}\right]$ and $\hat{\mathbf{j}} = \left[\begin{smallmatrix}1\\0\end{smallmatrix}\right]$ in the 2 dimensional euclidean space which can be thought of one unit of distance in the x and y direction respectively.

A linear combination of x and y is $ax + by$ where a, b are constants.

With this in mind, another way of writing a vector $\vec{a} = \left[\begin{smallmatrix}x\\y\end{smallmatrix}\right]$ is $\vec{a} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}}$. Notice that like in Definition 2.2, $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ are written in a linear combination of x and y. Of course, these are not the only basis vectors in 2d, if you have learned polar coordinates before, there is the equivalent $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$, but this is a topic for a later chapter. Next, we shall look at the *addition* of vectors. A vector $\vec{a} = \left[\begin{smallmatrix}x_a\\y_a\end{smallmatrix}\right]$ added to a vector $\vec{b} = \left[\begin{smallmatrix}x_b\\y_b\end{smallmatrix}\right]$ is the vector $\left[\begin{smallmatrix}x_a+x_b\\y_a+y_b\end{smallmatrix}\right]$. As you can see, the addition of two vectors results in another vector. Here is a illustration to help clear up things:

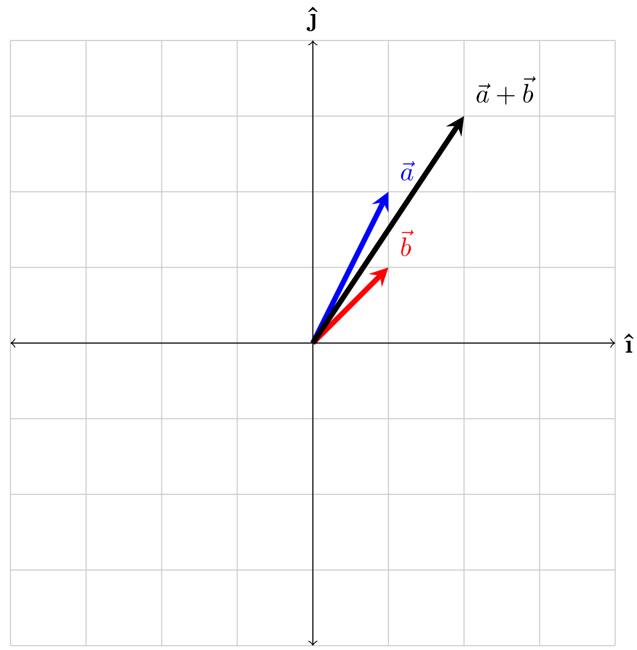


Figure 2.2. Illustration of vector addition.

Let us analyze Figure 2.2. Firstly, notice that $\vec{a} = 1\hat{\mathbf{i}} + 2\hat{\mathbf{j}}$ and $\vec{b} = 1\hat{\mathbf{i}} + 1\hat{\mathbf{j}}$. Thus, $\vec{a} + \vec{b} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$, exactly like in our figure. One way to understand this is to "attach" the vector $\vec{a}$'s tail to $\vec{b}$'s head by translating $\vec{a}$. You may have seen this in your physics class before. Similarly, the subtraction of vectors $\vec{x} = \left[\begin{smallmatrix}a\\b\end{smallmatrix}\right]$ and $\vec{y} = \left[\begin{smallmatrix}c\\d\end{smallmatrix}\right]$ is $\vec{x} - \vec{y} = \left[\begin{smallmatrix}a-c\\b-d\end{smallmatrix}\right]$. Of course, we can use the same "head to tail" model to explain vector subtraction. Subtracting $\vec{y}$ from $\vec{x}$ is simply finding the vector whose tail is on the head of $\vec{y}$ and whose head is on the head of $\vec{x}$. However, some of you may be scratching your heads right now, what if a vector does not start from the *origin* (or the coordinate $(0, 0)$) but its tail lands on, say $(1, 1)$. How do we express such a vector? The answer may be surprising, both vectors have the same representation. However, unless specifically stated, *all* vectors start from the origin. This is also why the "head to tail" model works, while the "tail" of the vector is not at the origin, translating it to the origin results in an *identical* vector.

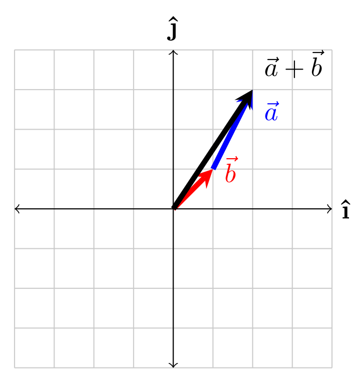


Figure 2.3. Illustration of vector "head to tail" addition.

Translation means moving something without rotation or any scaling.

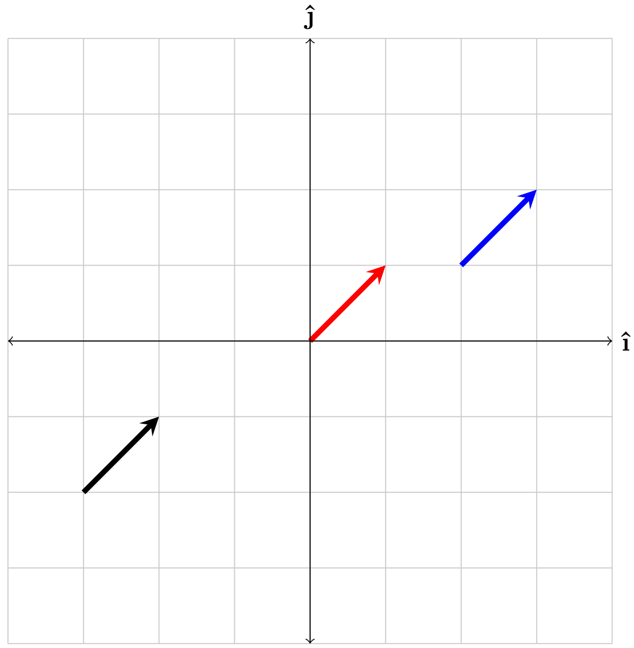


Figure 2.4. Three identical vectors.

Example Here are some examples of vectors:

$$\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \; \vec{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix} \; \vec{c} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

Here are some examples of adding and subtracting vectors. Note that these operations are *commutative*, in other words the order does not matter:

$$\vec{a} + \vec{b} = \begin{bmatrix} 1-3 \\ 2+4 \end{bmatrix} = \begin{bmatrix} -2 \\ 6 \end{bmatrix}$$

$$\vec{a} - \vec{c} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

$$\vec{a} + \vec{b} - \vec{c} = \begin{bmatrix} -2 \\ 6 \end{bmatrix} - \vec{c} = \begin{bmatrix} -4 \\ 8 \end{bmatrix}$$

Given a vector $\vec{a}$, $-\vec{a}$ can be thought of the same vector with the same magnitude but pointing in the *opposite* direction. To find it, simply flip the signs of all the components in $\vec{a}$. If you add $\vec{a}$ to $-\vec{a}$, you get what is called the *zero vector*, or the vector $\left[\begin{smallmatrix} 0 \\ 0 \end{smallmatrix}\right]$. This is a vector without direction has zero magnitude. We shall denote it $\vec{0}$.

Lastly, I want to briefly touch on the idea of *magnitude* which is basically the length of a vector. The derivation of the following formula is simply the Pythagorean theorem.

Definition 2.3 The magnitude of a vector $\vec{v}$ in n-dimensional space is as such:

$$||\vec{v}|| = \sqrt{v_1^2 + v_2^2 + ... + v_n^2}$$

A *unit* vector is a vector with magnitude 0.

Exercise

1. Given that $\vec{a} = \left[\begin{smallmatrix}3\\4\end{smallmatrix}\right]$ and $\vec{b} = \left[\begin{smallmatrix}-2\\1\end{smallmatrix}\right]$ and $\vec{c} = \left[\begin{smallmatrix}-3\\-2\end{smallmatrix}\right]$, calculate:
   (a) $\vec{a} + \vec{b}$
   (b) $\vec{a} - \vec{c}$
   (c) $\vec{a} - \vec{b} - \vec{c}$
2. Are the following statements true or false:
   (a) The subtraction of two vectors *always* results in a vector.
   (b) When any vector is added to the zero vector, the result *always* is not a zero vector.
3. Find the magnitude of $\begin{bmatrix}3\\4\end{bmatrix}$ and $\begin{bmatrix}3\\4\\5\end{bmatrix}$.
4. **(Difficult)** Prove that the addition of vectors is *associative* given that the addition in $\mathbb{R}$ is associative.

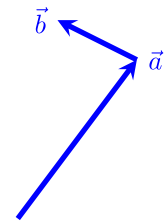


Figure 2.5. Visualisation of "head to tail" addition of $\vec{a}$ and $\vec{b}$.

An operation $\circ$ is associative if order in which it is applied does not matter, in this case $(\vec{x}+\vec{y})+\vec{z} = \vec{x}+(\vec{y}+\vec{z})$.

Solution

1. (a) $\begin{bmatrix}1\\5\end{bmatrix}$
   (b) $\begin{bmatrix}6\\6\end{bmatrix}$
   (c) $\begin{bmatrix}8\\7\end{bmatrix}$
2. (a) True since subtraction can be viewed as $\vec{a} + (-\vec{b})$ and $-\vec{b}$ is a vector.
   (b) False since $\vec{0} + \vec{0} = \vec{0}$.
3. 5 and $\sqrt{50}$.
4. Let $\vec{x}, \vec{y}, \vec{z}$ be vectors in $\mathbb{R}^n$ (N-dimensional euclidean space). Notice that:

$$\begin{aligned}
(\vec{x}+\vec{y})+\vec{z} &= \left(\begin{bmatrix}x_1\\ \dots\\ x_n\end{bmatrix} + \begin{bmatrix}y_1\\ \dots\\ y_n\end{bmatrix}\right) + \begin{bmatrix}z_1\\ \dots\\ z_n\end{bmatrix} \\
&= \begin{bmatrix}x_1+y_1+z_1\\ \dots\\ x_n+y_n+z_n\end{bmatrix} \\
&= \begin{bmatrix}x_1+(y_1+z_1)\\ \dots\\ x_n+(y_n+z_n)\end{bmatrix} \\
&= \begin{bmatrix}x_1\\ \dots\\ x_n\end{bmatrix} + \left(\begin{bmatrix}y_1\\ \dots\\ y_n\end{bmatrix} + \begin{bmatrix}z_1\\ \dots\\ z_n\end{bmatrix}\right) \\
&= \vec{x}+(\vec{y}+\vec{z})
\end{aligned}$$

SECTION 2.2

Scaling and dot product

In this chapter we will deal two kinds of multiplication with vectors. By far the easiest would be scaling a vector. As we have previously discussed, the magnitude of a vector is the length of a vector. But what if I want to "elongate" the vector by some amount to make it longer or shorter. This is called *scaling*.

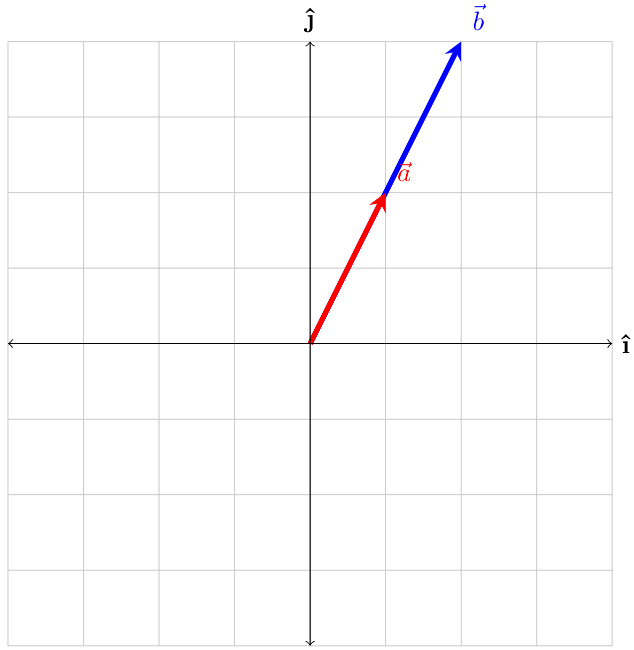


Figure 2.6. Two vectors, one twice as long as the other.

Take a look at Figure 2.6, $\vec{b}$ is twice as long as $\vec{a}$. What should we multiply to $\vec{a}$ to make as long as $\vec{b}$? If you guessed 2, the you are right. To multiply a vector by a constant, we multiply each individual component of the vector by the constant. From the figure, we can identify that $\vec{a} = \left[\begin{smallmatrix}1\\2\end{smallmatrix}\right]$ so $2\vec{a} = \left[\begin{smallmatrix}2\times1\\2\times2\end{smallmatrix}\right] = \left[\begin{smallmatrix}2\\4\end{smallmatrix}\right]$ which is equal to $\vec{b}$. You may extrapolate this to higher dimensions.

Definition 2.4

Scaling a n-dimensional vector $\vec{v}$ by a constant n is defined as such:

$$n\vec{v} = n\begin{bmatrix} v_1 \\ \dots \\ v_n \end{bmatrix} = \begin{bmatrix} n\times v_1 \\ \dots \\ n\times v_n \end{bmatrix}$$

Let us try something fun, what if we multiplied the vector by a negative number? Well, the computation remains the same. As you can see from Figure 2.7, $\vec{a} = (-0.5)\vec{b}$. You may wish to compute this yourself to verify. One thing to note here is that the vector gets flipped across the origin when the constant the vector is multiplied to is negative. Also, when scaling a vector by some constant, the reader may want to verify using Definition 2.3 that the magnitude of the vector is also scaled by that constant. This should be a trivial proof.

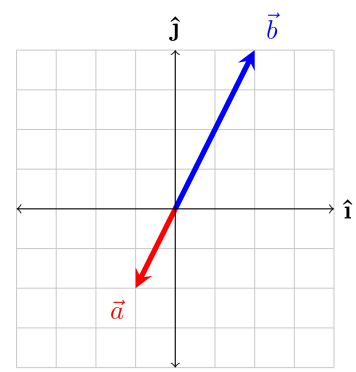


Figure 2.7. Two vectors, one is scaled by -0.5.

You may have noticed that I have abstained from using either the $\cdot$ or $\times$ operators. The reason is bad notation which is sprinkled throughout mathematics. These two operators have special meanings in the realm of linear algebra. We shall first explore the easier one, the *dot product*.

Dot products use the $\cdot$ operator and only acts on vectors, do not abuse notation by using the $\cdot$ operator in the context of scaling a vector. Dot products are actually fairly straightforward to compute:

Definition 2.5 The dot product of two vectors in n-dimensional space $\vec{a}$ and $\vec{b}$ is:

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n = (||\vec{a}||)(||\vec{b}||) \cos\theta$$

where θ is the angle between the two vectors. We shall refer to the formula between the first and second equal signs the "first definition of dot products" and the formula after the second equal sign the "second definition of dot products".

From Definition 2.5, we can tell that the dot product tells us something about the relationship between the two vectors.

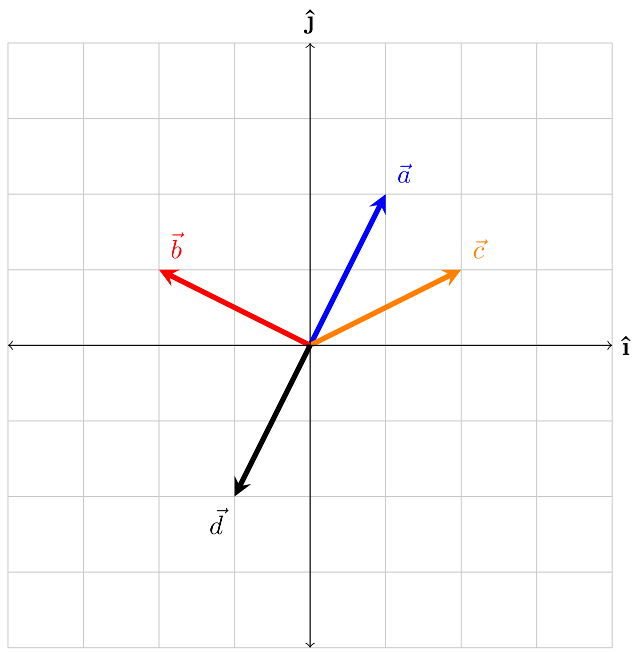


Figure 2.8. Four vectors pointing in different directions.

Firstly, note down the components of each vector in Figure 2.8. Then, let us compute $\vec{a} \cdot \vec{b}$. By Definition 2.5, it is $1(-2) + 2(1) = 0$. Also, it is important to take note that these two vectors are orthogonal(at right angles with each other). This brings us to an important property of dot products, if two vectors are orthogonal, then their dot product is 0. This can be seen from the fact that $\cos\left(\frac{\pi}{2}\right) = 0$ and from the second definition of dot products in Definition 2.5. In a sense, the dot product tells you how aligned are the directions of two vectors. Another example would be taking the dot product of two identical vectors, such as $\vec{a} \cdot \vec{a}$. The dot product there is $1(1) + 2(2) = 5$. In addition, we know $\cos 0 = 1$ so according to the second definition, $\vec{a} \cdot \vec{a} = (||\vec{a}||)(||\vec{a}||) = (||\vec{a}||)^2$. If we take the square root on both sides, $||\vec{a}|| = \sqrt{\vec{a} \cdot \vec{a}}$ which can be rewritten as $||\vec{a}|| = \sqrt{a_1^2 + \cdots + a_n^n}$. This is exactly the formula we got in Definition 2.3 for the magnitude of a vector. This is a pretty nice result.

Example

From Figure 2.8, we can see that:

$$\vec{a} \cdot \vec{c} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 1(2) + 2(1) = 4$$

$$\vec{a} \cdot \vec{d} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ -2 \end{bmatrix} = 1(-1) + 2(-2) = -5$$

Remark

Generally, when finding the dot product, we use the first form in Definition 2.5 as it is typically easier to compute. It is also important to know that the dot product of two vectors results in a number, not a vector like in scaling.

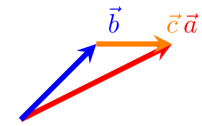


Figure 2.9. Figure of a triangle formed by three vectors.

Exercise

1. If $\vec{x} = \left[\begin{smallmatrix} 1 \\ 2 \end{smallmatrix}\right]$ and $\vec{b} = \left[\begin{smallmatrix} 3 \\ 4 \end{smallmatrix}\right]$, compute the following:
   (a) $2\vec{x}$
   (b) $\vec{x} \cdot \vec{y}$
   (c) $\vec{y} \cdot \vec{x}$
2. Prove or disprove that the dot product *commutes*.
3. Given two vectors $\vec{a}$ and $\vec{b}$ separated by an angle θ, they form a triangle with a third side $\vec{c} = \vec{a} - \vec{b}$ (See Figure 2.9).By taking the dot product of $\vec{c}$ and itself, derive the law of cosines. (Hint: $(\vec{x} \cdot \vec{y}) \cdot (\vec{z} \cdot \vec{w}) = \vec{x} \cdot \vec{z} + \vec{x} \cdot \vec{w} + \vec{y} \cdot \vec{z} + \vec{y} \cdot \vec{w}$)

An operation $\circ$ is commutative if $x \circ y = y \circ x$.

Solution

1. (a) $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$
   (b) 11
   (c) 11
2. The dot product commutes. From the second definition of the dot product from Definition 2.5, we can see that $\vec{a} \cdot \vec{b} = (||\vec{a}||)(||\vec{b}||)\cos(\theta) = (||\vec{b}||)(||\vec{a}||)\cos(\theta) = \vec{b} \cdot \vec{a}$. Therefore, the dot product commutes.
3. Let a, b, c be the magnitudes of the respective vectors. Just take the dot product of $\vec{c}$ and itself.

$$\begin{aligned} \vec{c} \cdot \vec{c} &= \left(\vec{a} - \vec{b}\right) \cdot \left(\vec{a} - \vec{b}\right) \\ &= a^2 - 2\vec{a} \cdot \vec{b} + b^2 \\ &= a^2 + b^2 - 2ab\cos(\theta) \end{aligned}$$

Which is exactly the law of cosines.

Note that $\vec{a} \cdot \vec{a} = (||\vec{a}||)^2$ as evident from the second definition of Definition 2.5.

Section 2.3
Matrices

In this chapter, we will introduce a new mathematical object that you have probably heard before but not sure what it is. To be honest, many mathematics students only know how to *compute* matrix operations but does not know the meaning behind them. I will try to do my best to explain them in a clear manner and I hope that you would appreciate matrices more for their beautiful geometric meaning than their mind-numbing calculations. A great place to start is the *basis vectors* introduced back in Definition 2.2. All vectors can be written as a unique linear combination of the basis vectors. But $\hat{\mathbf{\imath}}$ and $\hat{\mathbf{\jmath}}$ are not the only basis vectors out there, what if I wanted to describe vectors using another set of basis vectors? Perhaps that would be more convenient for my special use case. Well, it is actually pretty easy to do so. For the sake of this argument, let us try to rotate all of the vectors in a normal 2d vector field anti-clockwise by 90°. Notice how we could not describe $\vec{b}$ as some constant multiplied to $\vec{a}$ in Figure 2.10. We need some other tool to help us. If we change the basis vectors of $\vec{a}$ to another pair of basis vectors, can we somehow get $\vec{b}$? Here is a thought: replace the $\hat{\mathbf{\imath}}$ with $\left[\begin{smallmatrix}0\\1\end{smallmatrix}\right]$ and $\hat{\mathbf{\jmath}}$ with $\left[\begin{smallmatrix}-1\\0\end{smallmatrix}\right]$. So, $\vec{a} = 1\hat{\mathbf{\imath}} + 2\hat{\mathbf{\jmath}}$ will become $\vec{a} = \left[\begin{smallmatrix}0\\1\end{smallmatrix}\right] + 2\left[\begin{smallmatrix}-1\\0\end{smallmatrix}\right] = \left[\begin{smallmatrix}-2\\1\end{smallmatrix}\right]$. This is pretty intriguing, by changing the basis vectors, we have basically "morphed" $\vec{a}$ into $\vec{b}$. This "morphing" is called a *linear transformation* or *linear map*.

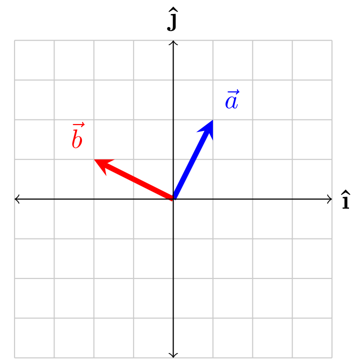


Figure 2.10. Two vectors, one rotated anti-clockwise by 90°.